

AMNA KHAN

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EDUCATION

Dr. A.P.J. Abdul Kalam Technical University

Dec 2020 – Jun 2024

Bachelor of Technology in Computer Science & Engineering | CGPA: 8.06/10

Uttar Pradesh, India

- Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, OOP, Software Engineering

EXPERIENCE

Application Development Engineer-I

Sep 2024 – Present

NCR Atleos

India

- Architected and shipped **Titan Dashboard**, a full-stack React/TypeScript platform serving finance teams across **35+ countries**, reducing manual batch tracking effort by **80%** and improving end-to-end processing performance by **40%**.
- Owned end-to-end development of **Oracle R12 enterprise applications** on OCI cloud and on-premises infrastructure, maintaining **99.9% uptime SLA** for mission-critical financial systems.
- Designed and automated **File Transfer System (FTS)**, reducing data processing latency by **60%**; integrated **IAM with RBAC** for Titan and Trident, securing access for **200+ users**.
- Engineered dynamic **database configuration pipelines** for BOC applications, cutting deployment failures by **35%** via automated schema validation and rollback mechanisms.
- Led cross-functional **RevPro revenue recognition testing** across business units, achieving **100% ASC 606 compliance** with zero audit findings.

Software Engineer Intern

Feb 2024 – Sep 2024

NCR Atleos

India

- Built **Titan Dashboard MVP** from scratch using React, Node.js, and PostgreSQL with WebSocket-based live updates, improving batch movement visibility by **50%** across **3 business units**.
- Implemented data validation and multi-level approval workflows, achieving **99.5% data accuracy** across financial records.
- Created **ERP Wiki**, consolidating technical documentation for **5 enterprise modules** (Cloud, Treasury, Legacy Apps, Planning, Forecasting).

PROJECTS

NexusAI – Document Intelligence Platform | *Full-Stack AI*

Next.js, FastAPI, Ollama, ChromaDB, PostgreSQL

- Architected production-grade **RAG platform** enabling natural language Q&A over enterprise documents with **94% answer accuracy** across **1,200+ document chunks**, using Ollama Llama 3 for LLM generation and recursive document chunking.
- Engineered semantic search pipeline with **ChromaDB** (768-dim vector embeddings), async task processing via **Celery + Redis**, PostgreSQL metadata storage, and source citation tracking. Fully self-hosted – **zero external API costs**.

SyncBoard – Real-Time Collaborative Workspace | *SaaS*

Next.js 14, TypeScript, Prisma, Redis, Socket.io

- Scaled real-time collaborative SaaS to **50+ concurrent users** with Kanban boards, live cursors, and team chat at **100ms sync latency** via WebSocket + Redis pub/sub with **optimistic UI updates** (Zustand state management).
- Implemented **NextAuth** (OAuth + JWT) authentication, Prisma ORM with PostgreSQL, and **role-based access control** with row-level security policies for multi-tenant isolation.

Sentinel – Infrastructure Monitoring | *Distributed Systems*

Node.js, Kafka, TimescaleDB, Redis, WebSocket

- Designed fault-tolerant monitoring for **26 microservices** with **Kafka event streaming at 12.4k req/s**, TimescaleDB time-series aggregation, latency percentile dashboards (p50/p95/p99), and **anomaly detection**.
- Built Next.js observability dashboard with real-time health grid, throughput charts, and **automated incident escalation** via WebSocket push notifications.

Pathfinding Visualizer | **WeMeet – Video Conferencing**

React.js, JavaScript, Agora WebRTC

- Built interactive algorithm visualizer (Dijkstra's, A*, BFS, DFS) with real-time grid rendering and weighted nodes; **WeMeet**: HD video conferencing with **Agora RTC/RTM**, adaptive bandwidth optimization, and sub-200ms latency.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, C/C++, SQL, HTML/CSS

Frontend: React 18, Next.js 14, Tailwind CSS, Framer Motion, Three.js, Zustand

Backend: Node.js, FastAPI, Express.js, Prisma ORM, SQLAlchemy, WebSocket, Socket.io, REST APIs, Celery

Data & AI: PostgreSQL, MongoDB, Redis, TimescaleDB, ChromaDB, Kafka, Ollama (Llama 3), RAG Architecture

Cloud & DevOps: AWS (EC2, S3), OCI, Docker, CI/CD Pipelines, Git/GitHub, Nginx, Linux

Testing & Methods: Jest, Pytest, Postman, Jira, Agile/Scrum, System Design, Code Review, Technical Documentation

ACHIEVEMENTS

- Solved **500+ DSA problems** across LeetCode & HackerRank; achieved **6-star Problem Solving** and **5-star C++ Programming** ratings on HackerRank.
- Shipped **production systems at NCR Atleos** (Titan Dashboard, FTS, ERP Wiki) actively used by finance teams across **35+ countries** in global operations.
- Built **3 production-grade full-stack applications** with system design documentation, covering **AI/ML pipelines**, **real-time collaboration**, and **distributed infrastructure monitoring**.